Code: DMS-02 **Paper ID: 13552**

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)

(TRIMESTER –III) END TERM EXAMINATION, MAY -2025

SUPPLY CHAIN NETWORK DESIGN

[Time: 2 Hours] ***[Maximum Marks: 60]***

Following are the course outcomes of the subject: - Supply Chain Network Design

CO	Course Outcome (CO)
CO1	Understand the foundational concepts and principles of supply chain networks, emphasizing their significance in modern business operations.
CO2	Analyze the diverse factors influencing network design decisions, such as cost, service levels, and agility, to develop a holistic perspective on network management.
CO3	Identify the key factors to be considered when designing a distribution network

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)

Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	**4**	CO - 1	BL 1
(B)	**4**	CO - 1	BL 2
(C)	**4**	CO - 1	BL 1
(D)	**4**	CO - 1	BL 1
(E)	**4**	CO - 1	BL 1

SECTION-B

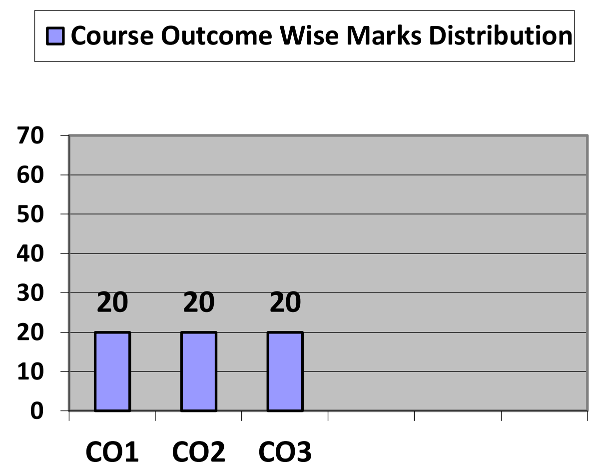
Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)

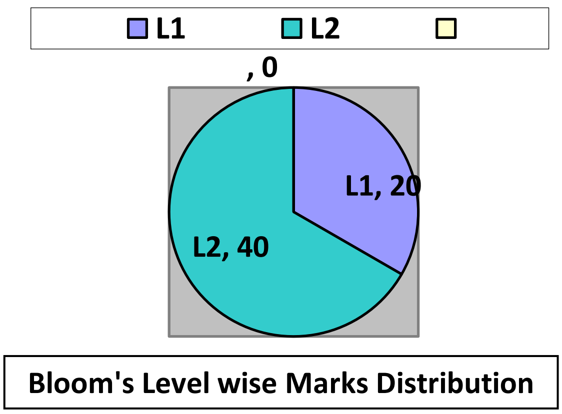
Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	**10**	CO – 2	BL 2
2 (B)	**10**	CO – 2	BL 2
3 (A)	**10**	CO – 3	BL 2
3 (B)	**10**	CO – 3	BL 2

SECTION-C

Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	**10**	CO – 2	BL 2
4 (B)	**10**	CO - 3	BL 2





Moderated By **IQAC** **Dean - Examinations**

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SUPPLY CHAIN NETWORK DESIGN

[Time: 2 Hours] ***[Maximum Marks: 60]***

INSTRUCTIONS:
(i) There are THREE sections in this question paper.
(ii) Attempt questions from each section as per the directions.
(iii) Make suitable assumptions wherever necessary.

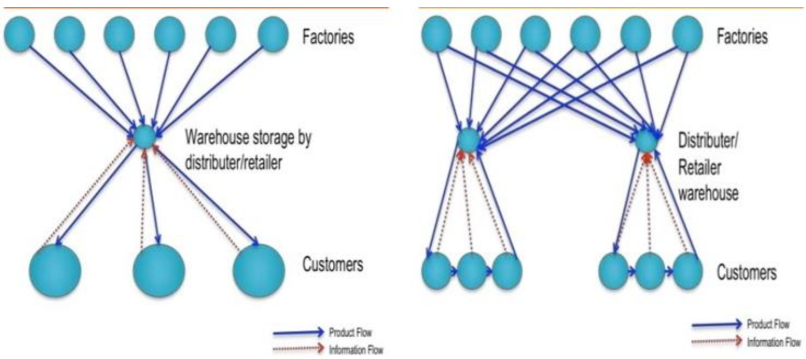
SECTION-A
(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following: ***(5×4=20)***

a) Similar kind of business can opt different kind of networks, either centralised or Decentralised. Do you Agree? Define with reasons.
b) There are several Solid factors in Designing a Supply Chain Network. Explain at least 5 factors.
c) The Cost and Response time has inverse relationship. Do you agree? Define with reasons.
d) Global Supply Chain Networks are need of the hour. Do you agree? Define with reasons.
e) Which of the network have the following advantages, explain with reasons:
   a) Better Responsiveness b) Better Structure & Control
   c) Better Flexibility & Scalability

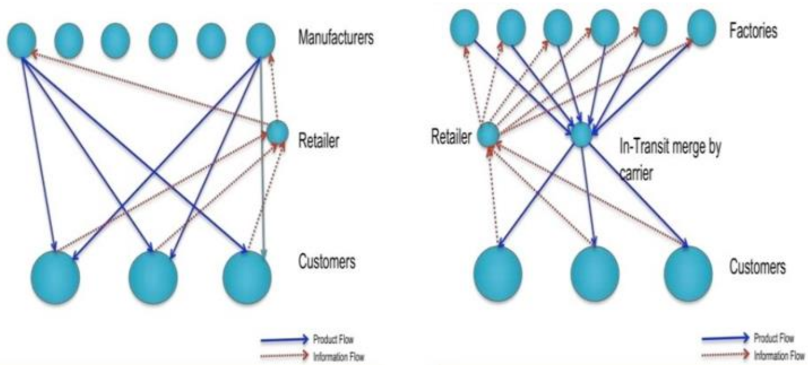
SECTION-B
(Long Answer Type Questions)

Q.2. Attempt all questions of the following ***(2X10=20)***



(a) Identify the networks and explain their Cost and Service Factors.

OR



(b) Identify the networks and explain their Cost and Service Factors.

Q.3. *(a)* Organizations often follow a multi-phase approach when making strategic supply chain network design decisions. Illustrate how a company might approach this process in stages from defining supply chain strategy to selecting specific facility locations. In your answer, highlight the types of internal and external factors that could influence decisions at each stage.

OR

(b) In the context of supply chain network design, companies often face strategic decisions between offshoring, onshoring, and outsourcing. Outline the key factors that influence these decisions and analyse how each approach can impact cost efficiency, flexibility, and risk in the supply chain. Provide examples to support your analysis.

SECTION-C

Case Study: NovaTech Electronics – A Holistic Approach to Global Supply Chain Network Design

NovaTech Electronics, a fast-growing consumer electronics company headquartered in Munich, Germany, is reaching a turning point in its global strategy. Known for its smart home devices, wearable technology, and compact entertainment systems, the company has seen consistent success across Europe. However, its recent expansion into North America, Southeast Asia, and South America is exposing limitations in its current supply chain model. At present, NovaTech relies on a centralized manufacturing facility in Eastern Europe and a regional distribution center in the Netherlands. While this setup has worked well within Europe, it has led to increasing delivery delays, higher transportation costs, and reduced service levels in newer markets. As competition intensifies and customer expectations grow, especially around delivery speed and availability, the company realizes it must redesign its global supply chain network. To support its expansion and meet diverse market needs, NovaTech is considering shifting part of its production to lower-cost regions such as Vietnam or India. This offshoring strategy could reduce operational expenses but may introduce risks related to political instability, customs delays, and long transit times. As an alternative, the company is also exploring the idea of moving certain final assembly operations closer to key demand hubs like the United States. This onshoring approach could improve service levels and responsiveness but might come at a higher cost. Alongside these considerations, NovaTech is weighing the benefits of outsourcing some of its non-core operations, such as logistics and warehousing, to third-party providers with specialized expertise. While outsourcing could help the company focus more on innovation and core competencies, it would also require careful coordination and trust in external partners.

The leadership team is evaluating a complex mix of factors to guide their decisions. These include the total cost of operations across the supply chain, customer service expectations in different markets, and the need for agility in responding to market shifts, demand fluctuations, or supply disruptions. They are also looking at the availability and quality of infrastructure in potential locations, the reliability of transportation networks, and the digital capabilities required to manage a distributed supply chain. Regulatory conditions, such as import-export restrictions and compliance requirements in various countries, are also critical to consider. A key focus of the redesign is the structure of NovaTech's distribution network. The company is questioning whether its current centralized model is still suitable as it expands globally. It is contemplating the establishment of regional distribution centers in North America, Southeast Asia, and South America. While this decentralized approach could improve delivery speeds and enhance customer satisfaction, it could also increase operational complexity and coordination efforts. Ultimately, NovaTech aims to create a global supply chain network that balances cost-efficiency, high service levels, and operational flexibility. The leadership team understands that prioritizing only one dimension, such as cost, may compromise the others. A network optimized purely for low cost may lack the agility needed to deal with today's unpredictable global environment. On the other hand, a highly agile and service-oriented network might result in significantly higher costs if not managed carefully. The challenge lies in designing a network that supports growth and resilience without sacrificing efficiency.

Q.4. Attempt all parts of the following: ***(2×10=20)***

4(a) Based on NovaTech's situation, explain how cost, service levels, and agility influence the decisions involved in designing a global supply chain network. Use examples from the case to illustrate how these elements interact and potentially conflict with one another.

4(b) Considering NovaTech's plans to establish new distribution centers, classify the key strategic and operational factors the company should consider when designing its distribution network. Reflect on how these choices could impact the efficiency, responsiveness, and overall performance of the supply chain.